

2019-Ph-H1-Q7

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

Summary:

The terminal velocity v_t of a skydiver is given by:

$$v_t = \sqrt{\frac{2mg}{\rho AC_d}}.$$

We are to find the **area** A of a skydiver with:

$m = 95 \text{ kg}$, $g = 9.8 \text{ N/kg}$, $C_d = 1.0$, $\rho = 1.21 \text{ kg/m}^3$, $v_t = 44 \text{ m/s}$.

Solution:

Rearrange for A :

$$A = \frac{2mg}{\rho C_d v_t^2}.$$

Substitute values:

$$A = \frac{2 \times 95 \times 9.8}{1.21 \times 1.0 \times 44^2} = \frac{1862}{1.21 \times 1936} = \frac{1862}{2344.56} \approx 0.79 \text{ m}^2.$$

Answer: B. 0.79 m^2

Guidance for Students:

- Always square the velocity when using this relationship.
- Check units: mass (kg), g (N/kg), area (m^2), ρ (kg/m^3).

Revision Tips:

- **Rearrange equations** carefully before substituting.
- A good estimate for area helps confirm your calculation (around 1 m^2 for a human).
- Terminal velocity depends on both **surface area** and **drag coefficient**.

